

Automatic Composite-Modulation Classification Using Ultra Lightweight Deep-Learning Network Based on Cyclic-Paw-Print

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Abstract—Automatic composite-modulation classification (ACMC) has been considered as an essential function in the next generation intelligent telemetry, tracking & command (TT&C), cognitive space communications, and space surveillance. This paper introduces a novel ACMC scheme using the cyclic-paw-print extracted from the composite-modulation (CM) signals. In this new framework, the cyclic-spectrum analysis is first invoked to acquire the polyspectra of the received CM signals corrupted by different fading channels. Then, a new feature, namely cyclic-paw-print (CPP), is established upon the image representation of the cyclic spectrum, which can be robust against channel noise. Then, a highly-efficient ultra lightweight deep-learning network (ULWNet), which takes the CPPs as the input features, is designed to identify the composite modulation type. Our proposed new scheme can greatly improve the computational efficiencies incurred by the existing deep-learning networks and capture more reliable features latent in CM signals to result in an excellent classification accuracy. Monte Carlo simulation results demonstrate the effectiveness and the superiority of our proposed new ACMC scheme to the existing deep-learning networks.

Index Terms—Automatic composite modulation classification (ACMC), cyclic spectrum, cyclic-paw-print (CPP), ultra lightweight network (ULWNet), cognitive telemetry, tracking & command (TT&C), cognitive space communications.

I. INTRODUCTION

WITH the broad deployment of the fifth generation (5G) wireless networks, the design and planning of the sixth generation (6G) wireless communication networks have been appealing to the global academia and industry [1].

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Space-air-ground integrated network architectures and intelligent communications empowered by artificial intelligence are crucial to the future 6G wireless systems [2], [3], [4]. The incorporation of terrestrial, aerial, and satellite communication systems not only significantly increases the demands on frequency resources and the sophisticated operation of the network, but also results in the broad diversity of the transmitted signals which need to be automatically (blindly) recognized in intelligent communications. To improve spectral efficiency and avoid interference, automatic modulation classification (AMC) has been designed to identify the composite modulation schemes, which have been widely used in cognitive space communications and the next-generation intelligent tracking, telemetry, and command (TT&C) systems [5].

Composite modulation (CM), which is referred to as the “multi-layer modulation of multi-service baseband data using different modulation schemes”, has been widely adopted in almost all space missions to facilitate TT&C, terrestrial communications, and space sensing due to its outstanding capability of improving the transmission efficiency, mitigating the multi-user interference, and enhancing the communication security [6], [7], [8], [9], [10], [11]. As cognitive radio (CR) is incorporated into the next generation of cognitive TT&C and broadband space communications, the CM scheme of the signal transmitted by the primary user must be blindly identified by the secondary users without any *a priori* knowledge. Thus, *automatic composite modulation classification* (ACMC) is a very important mechanism for serving this purpose.

There exist two common AMC approaches, namely (i) the likelihood-based (LB) decision-theoretic approach (see [12], [13], [14]) and (ii) the feature-based (FB) pattern-recognition approach (see [15], [16], [17], [18]), which are under the assumption that the target signal is modulated by the single-layer modulation scheme. The existing AMC approaches lead to promising classification accuracies for such single-layer modulated signals. However, when directly applied to identify the CM scheme of an unknown received CM signal, the existing AMC schemes inevitably suffer from significant performance degradation due to the difficulty in extracting robust features in the presence of nonlinear coupling between multi-layer modulations. Moreover, the implementation of the existing AMC techniques in the resource-constrained scenarios, particularly in aerospace

applications, would be quite impractical due to their huge computational complexities.

To the best of our knowledge, very few ACMC techniques have been reported in the literature. Most existing ACMC methods belong to the FB approach. Features employed in the conventional AMC schemes have also been adopted for ACMC, including *fractional Fourier transform* in [19], *high-order cumulants* (HOCs) in [20], and *instantaneous frequency-histogram and cyclic-spectrum* in [21]. In [19], features were first extracted from the fractional Fourier transform of an unknown received CM radar signal, and then a decision tree was employed for ACMC. This scheme could lead to a promising classification performance for radar signals but its performance would dramatically degrade when this method is applied to the CM signals utilized in TT&C and space communications due to the mismatch of signal models. The ACMC method proposed in [20] extracted four characteristic parameters based on the HOC and the square spectrum of an unknown received CM signal, and then the classification was carried out using a decision tree. According to the simulation results shown in [20], the average classification accuracy for eight different CM schemes can reach up to 98% in the high SNR regime. However, its performance would dramatically degrade when the SNR slightly decreases, and the estimation variance is large when the sample size is small (generally speaking, we need to collect at least dozens of thousands of received signal data-samples to estimate a cumulant reliably as a rule of thumb). In [21], the main carrier modulation type of the received CM signal was identified according to the features generated from the instantaneous frequency-histogram and cyclic-spectrum of the signal. This method could lead to promising classification accuracies for the out-layer modulation recognition in low SNR conditions, while the complete CM scheme of an arbitrary target CM signal could not be identified. In addition to the aforementioned FB approach for ACMC, a two-fold phase-locked loop (PLL) and a fit test were also employed to identify the CM scheme in [22], where the PLL extracted the received CM signal's essential features and the goodness-of-fit test measured the distance between the phase cumulative distribution of the demodulated signal sequence and the ideal template distribution representing a particular CM scheme. Nonetheless, this method could not be effective when the PLL failed to lock, i.e., it would be very sensitive to the hardware capacity.

Compared to the classical AMC techniques, deep-learning (DL) networks can simplify the classifiers involved in the FB approach. DL networks have been applied recently for AMC of single-layer modulated signals [23], [24], [25], [26], [27], [28], [29], but they have not been applied for ACMC so far. A hybrid knowledge and data-driven deep learning framework has been adopted for AMC in [23]. The cyclo-stationary features built from the spectral correlation function (SCF) and the high-order statistics (HOS) of the received signal were fed into a convolutional neural network (CNN) to conduct AMC in [24]. The fixed boundary-range based empirical wavelet transform (FBREWT) was also invoked to extract the modulation features as the input of the DL network in [25]. A data-driven subsampling strategy based

on deep neural network architectures was designed for AMC in [26], which could reduce the training time and improve the classification accuracy simultaneously. Besides, residual neural networks (ResNets) have also been adopted for AMC: the optimized ResNet proposed in [27] and the ResNet in conjunction with the bi-objective memetic algorithm proposed in [28]. The long short-term memory (LSTM) with the global feature learning capability was adopted in [29] to identify the modulation scheme of the received signal based on the corresponding in-phase/quadrature-phase (I/Q) features. To maintain low computational complexities, pruning techniques and lightweight networks have been adopted for AMC [30], [31], [32], [33]. A pruning technique based on activation maximization (AM) was proposed in [30], which omitted the less important convolutional filters to reduce the network complexity. Various lightweight network architectures, including the multi-scale (MS) network in [31], the decentralized learning AMC (DecentAMC) network in [32], and the LightAMC in [33], were designed to achieve the promising accuracies with relatively small computing-resource consumption. Since the ACMC techniques for LEO (low earth-orbit) satellite communication networks and broadband space communications are necessary by the resource-constrained payloads of the spacecrafts in aerospace, the computational complexity of a future DL-based ACMC mechanism should be as low as possible. Hence, new robust and computationally-efficient ACMC schemes are surely in urgent demand for enhancing the performances of the current LEO satellite communication networks and space communication systems.

According to our observation of the CM signals in the cyclic-spectrum domain, we have found that the planform images of the *second-order cyclic spectra* (SCSs) of different CM signals demonstrate distinguishable features. Moreover, such robust features, i.e., the aforementioned planform images, are very robust against the CM signal distortions due to transmission channels. We name such a planform image “*cyclic-paw-print* (CPP)” as it looks like a collection of four symmetric paw prints. Motivated by this new discovery, we introduce a new ACMC method, which adopts the CPPs as the input features and our proposed novel “*ultra lightweight network* (ULWNet)” as the classifier to identify the CM scheme in cognitive TT&C or broadband space communications. The ULWNet is first trained to meet the resource-constrained requirement in aerospace applications to recognize the CM schemes based on the extracted CPPs. Then in the test stage, the CPP of an unknown CM signal is extracted from the second-order cyclic spectrum (SCS) of the signal. The classifier, namely the trained ULWNet, will be able to identify the CM scheme of this unknown signal using the corresponding CPP.

Finally, Monte Carlo simulations are carried out to test the effectiveness of our proposed new DL network, ULWNet, in comparison with other popular DL networks for ACMC. The simulation results demonstrate that the probability of correct classification P_{cc} of our proposed novel ACMC method can lead to as high as 99.8% with a remarkably low computational complexity when the SNR is as low as 0dB. The primary contributions of this work can be summarized as follows:

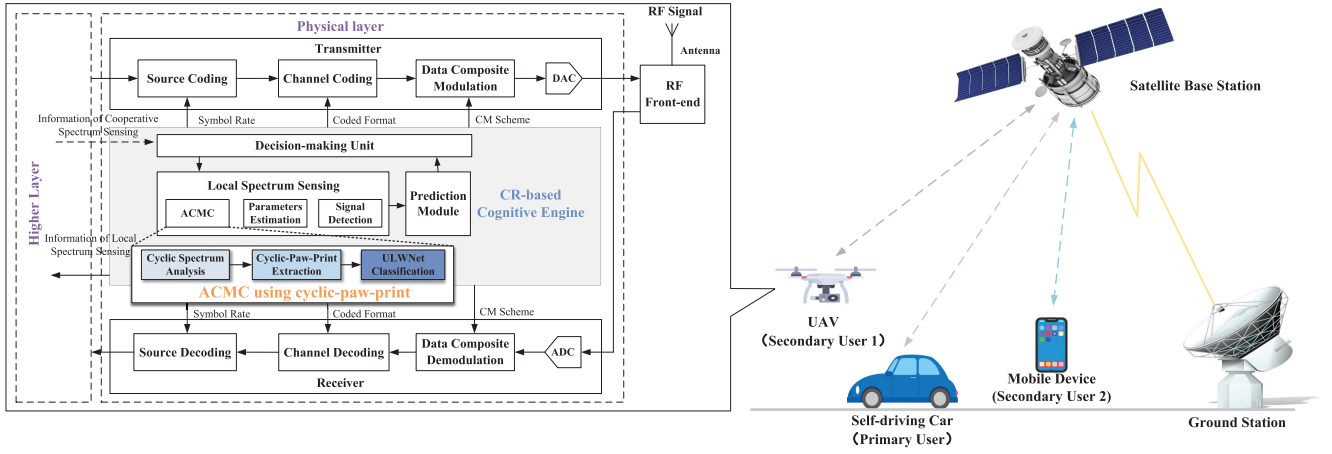


Fig. 1. A conjectured 6G communication system based on cognitive radio.

- A new modulation feature of the CM signals, named the cyclic-paw-print (CPP) which is the top view of the discrete normalized second-order cyclic spectrum (DNSCS) of the received signal, is proposed here.
- The digested (concise) CPP-based feature construction mechanism is proposed to transform the colormap of the magnitudes of a DNSCS into a fixed-size square pseudo-color image represented by a third-order tensor.
- To accommodate the practical resource-constrained applications, a highly-efficient ultra lightweight deep-learning network (ULWNet) with a fairly low computational complexity is designed to serve as the classifier for ACMC.
- The study in this work shows that the CPPs produced from CM signals are robust against various types of channels for ACMC and the ULWNet using CPPs leads to the promising recognition performance.

This paper is organized as follows. Section II introduces the CM system model and transmission model of CM signals. Section III presents the details of our proposed novel ACMC approach using the new ULWNet. Monte Carlo simulation results for ACMC are presented in Section IV, where the proposed ULWNet is compared with the other DL models and the HOS (high-order statistics) scheme. Finally, conclusion will be drawn in Section V.

Nomenclature: Scalars are denoted by Italianized alphabets such as a ; vectors are denoted by alphabets with overhead arrow notions such as $\vec{A}$; matrices are denoted by alphabets with overhead tilde notions such as $\tilde{A}$; sets are denoted by blackboard bold alphabets such as $\mathbb{A}$; tensors are denoted by bold alphabets such as $\mathbf{A}$. $\vec{A}^T$ and $\tilde{A}^T$ represent the transposes of a vector $\vec{A}$ and a matrix $\tilde{A}$, respectively. The size of a square matrix is represented by the total number of its columns (or rows). The set of all real numbers is denoted by $\mathbb{R}$, and $j \stackrel{\text{def}}{=} \sqrt{-1}$. $\Re\{\}$ and $\Im\{\}$ return the real and imaginary parts of the complex number enclosed by the braces, respectively.

II. SYSTEM MODEL

CM is undertaken by modulating a primary carrier in conjunction with multiple subcarriers onto which multi-user data

are multiplexed, and it is commonly employed by aerospacecraft for telemetry, telecommand, and space communications. In the future, a 6G communication network would facilitate the complete air-ground integration of heterogeneous communication technologies. A conjectured 6G communication system based on CR incorporated with ACMC is illustrated by Fig. 1. It is generally expected that a 6G communication system is a space-air-ground integrated communication network based on CR [2], [3], [4]. An intelligent 6G transceiver can sense the environmental variations and adapt the system parameters such as coding and modulation schemes dynamically to such variations, thus achieving better spectrum utilization and improving communication performance. The physical layer of the intelligent 6G transceiver consists of a transmitter, a receiver, and a CR-based cognitive engine. The spectrum sensing mechanism in the CR-based cognitive engine continuously monitors the signal activities in the frequency band of interest and detects and captures the dominant signal in presence. Then, the received signal, which is first appropriately filtered and synchronized, is processed to estimate the inherent parameters, and the associated modulation scheme is further identified by our proposed new ACMC technique. The spectrum-allocation strategy and the configuration parameters of the transmitter and receiver are dynamically adjusted to realize cognitive communications according to the instantaneous sensed information. Our proposed novel DL-based ACMC method can play a critical part in such an intelligent 6G transceiver that can reliably identify the CM scheme of an unknown received signal.

According to the Consultative Committee for Space Data Systems (CCSDS) standard in [5], various combinations of modulation schemes have been adopted by the two-layer hybrid modulation architecture for different space data-transmission applications. For example, the unified S-band TT&C system utilizes binary phase-shift keying (BPSK) and phase modulation (PM) jointly to implement CM [5]. Therefore, two categories of double-layer CM strategies are investigated here, including “pulse code modulation (PCM)/ M -ary phase-shift keying (M -PSK)/phase modulation (PM)” and “pulse code modulation (PCM)/ M -ary phase-shift keying

(M -PSK)/frequency modulation (FM)", where M -PSK is adopted for the inner-layer modulation and the middlewares are further modulated using PM or FM.

Assume that there are $\mathcal{J}$ subcarriers in a 6G communication system, and each individual user's information data stream is modulated onto the i^{th} subcarrier, $i = 1, 2, \dots, \mathcal{J}$. The inner M -PSK modulated signal $s_i(t)$ for i^{th} subcarrier (user) can be expressed by

$$s_i(t) \stackrel{\text{def}}{=} \Re \left\{ s_i(t) e^{j2\pi(m_i-1)/M_i} e^{j(2\pi f_i t + \phi_{i,0})} \right\},$$

$$m_i = 1, 2, \dots, M_i, \quad 0 \leq t \leq T_i, \quad (1)$$

where M_i , T_i , and $\phi_{i,0}$ denote the modulation order, symbol period, and initial phase of the i^{th} subcarrier, respectively; f_i represents the frequency of the i^{th} subcarrier; the pulse train of the i^{th} subcarrier is given by

$$g_i(t) \stackrel{\text{def}}{=} \sum_{q=0}^N a_{i,q} g(t - qT_i); \quad (2)$$

N denotes the sample size (total number of information symbols contained in the received signal), $a_{i,q}$ denotes the information symbol of the i^{th} subcarrier in the q^{th} symbol period, and $g(t)$ specifies the NRZ-L pulse waveform as given by

$$g(t) \stackrel{\text{def}}{=} \begin{cases} 1, & 0 \leq t < T_i, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

According to [34], the symbol rate of the inner M -PSK modulation ranges from 7.8125 to 4000 symbols/second (Sps), and the subcarrier frequency f_i shall be an integer multiple of the corresponding symbol rate. Then, all inner-modulated signals, namely $s_i(t)$, $i = 1, 2, \dots, \mathcal{J}$, are subsequently modulated onto the primary carrier using PM or FM.

A transmitted passband PCM/ M -PSK/PM signal $s_{\text{PCM/PSK/PM}}(t)$ consisting of $\mathcal{J}$ subcarriers can be expressed by

$$s_{\text{PCM/PSK/PM}}(t) \stackrel{\text{def}}{=} \Re \left\{ x_{\text{B}}^{\text{PM}}(t) \exp [j2\pi f_p t + \Phi_0] \right\}, \quad (4)$$

where $x_{\text{B}}^{\text{PM}}(t)$ denotes the equivalent complex baseband signal as given by

$$x_{\text{B}}^{\text{PM}}(t) \stackrel{\text{def}}{=} A \exp \left[jK_p \sum_{i=1}^{\mathcal{J}} s_i(t) \right]; \quad (5)$$

A , f_p , and Φ_0 denote the amplitude, the primary carrier frequency, and the initial carrier phase, respectively; K_p represents the phase-modulation index for the outer-layer primary PM, and its normal range is $0.2 \leq K_p \leq 1.4$ according to [34]. Similarly, a transmitted passband PCM/ M -PSK/FM signal $s_{\text{PCM/PSK/FM}}(t)$ consisting of $\mathcal{J}$ subcarriers can be expressed by

$$s_{\text{PCM/PSK/FM}}(t) \stackrel{\text{def}}{=} \Re \left\{ x_{\text{B}}^{\text{FM}}(t) \exp [j2\pi f_p t + \Phi_0] \right\}, \quad (6)$$

where $x_{\text{B}}^{\text{FM}}(t)$ denotes the equivalent complex baseband signal as given by

$$x_{\text{B}}^{\text{FM}}(t) \stackrel{\text{def}}{=} A \exp \left[jK_f \int_0^t \sum_{i=1}^{\mathcal{J}} s_i(\tau) d\tau \right]; \quad (7)$$

K_f is the frequency-modulation index for the outer-layer primary FM, and its normal range is $0.2 \leq K_f \leq 1.4$ according to [34]. For each $s_i(t)$, the subcarrier waveform and the corresponding inherent data stream can be synchronized in time with the coincidental zero-crossings such that $\phi_{i,0} = 0$, $i = 1, 2, \dots, \mathcal{J}$.

For notational simplicity, these two types of CM signals formulated by Eqs. (4) and (6) and the corresponding equivalent complex baseband signals formulated by Eqs. (5) and (7) are both denoted by $s_{\text{CM}}(t)$ and $x_{\text{B}}(t)$, respectively, where $s_{\text{CM}}(t) \in \{s_{\text{PCM/PSK/PM}}(t), s_{\text{PCM/PSK/FM}}(t)\}$ and $x_{\text{B}}(t) \in \{x_{\text{B}}^{\text{PM}}(t), x_{\text{B}}^{\text{FM}}(t)\}$.

At the intelligent receiver, the received CM signal, which is distorted by the propagation channel often characterized as an additive white Gaussian noise (AWGN) channel in the presence of Doppler and phase noises, can be expressed by

$$r(t) \stackrel{\text{def}}{=} \Re \left\{ x_{\text{B}}(t) \exp [j\varphi_d + \varphi_p] + n(t) \right\}, \quad (8)$$

where φ_d denotes the carrier phase evolution associated with the Doppler dynamics; φ_p specifies the phase noise introduced by the propagation channel, which involves the initial carrier phase Φ_0 ; $n(t)$ represents the AWGN with zero mean and variance σ^2 . By considering the Doppler effect, we may express the received equivalent complex baseband signal $x_{\text{B}}(t)$ by

$$x_{\text{B}}(t) \stackrel{\text{def}}{=} \begin{cases} A \exp \left[jK_p \sum_{i=1}^{\mathcal{J}} s'_i(t) \right] & \text{for PCM/M - PSK/PM,} \\ A \exp \left[jK_f \int_0^t \sum_{i=1}^{\mathcal{J}} s'_i(\tau) d\tau \right] & \text{for PCM/M - PSK/FM,} \end{cases} \quad (9)$$

where

$$s'_i(t) \stackrel{\text{def}}{=} \Re \left\{ g_i(t) \exp \left\{ j2\pi [\ell'_i t + (m_i - 1)/M_i] \right\} \right\},$$

$$m_i = 1, 2, \dots, M_i, \quad 0 \leq t \leq T_i; \quad (10)$$

ℓ'_i represents the received i^{th} subcarrier frequency; $\ell'_i \stackrel{\text{def}}{=} \ell_i + \ell_d$ and ℓ_d denotes the Doppler frequency offset.

III. THE PROPOSED NEW ACMC APPROACH

The block diagram of the proposed new ACMC approach using our proposed ULWNet based on CPP is illustrated by Fig. 2, which is composed by three major components: (i) *cyclic-spectrum analysis*, (ii) *CPP extraction*, and (iii) *ULWNet*. According to Fig. 2, the cyclic-spectrum analysis is applied to the received CM signal to generate the corresponding three-dimensional SCS, which is further utilized to form the CPP matrix from the SCS. Finally, the novel ULWNet, which has been pre-trained, to identify the CM scheme of the received signal according to the CPP. The details of the aforementioned three major components of our proposed new ACMC approach are presented as follows.

A. Cyclic-Spectrum Analysis

According to [35], [36], [37], features extracted from the SCSs of the received communication signals lead to very promising AMC accuracies. Instead of encoding such SCSs

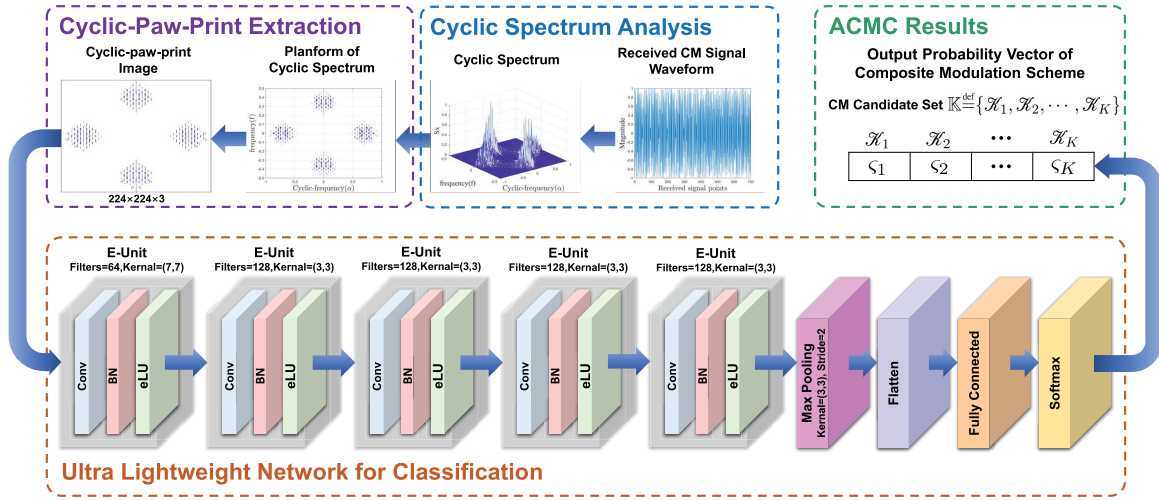


Fig. 2. The block diagram of our proposed new ACMC approach.

into graphs as [35], [36], [37], we propose to process them in an alternative way, i.e., project each SCS onto a grayscale colormap on the two-dimensional (spectral-frequency-cyclic-frequency) plane. In other words, each SCS is converted to the corresponding CPP tensor. The CPP tensors can be deemed regular color images obviously and they turn out to be the reliable features for ACMC. Given a received CM discrete-time signal sequence $r'(n)$ with the sample size N , the second-order cyclic autocorrelation function can be derived by utilizing the *FFT (fast Fourier transform) accumulation method (FAM)*. According to [38], [39], [40], [41], the discrete time-smooth-based cyclic periodogram $R_{r_T}^\alpha(n, f)_{\Delta t}$ for a spectral frequency f and a cyclic frequency α is given by

$$R_{r_T}^\alpha(n, f)_{\Delta t} \stackrel{\text{def}}{=} \frac{1}{ML} \sum_{\lambda=0}^{ML-1} r_T(\lambda, f_1) r_T^*(\lambda, f_2) g'(n - \lambda) e^{-j2\pi q \lambda / N}, \quad (11)$$

where $g'(n)$ is a unity-area weighting function of the width $\Delta \stackrel{\text{def}}{=} NT_s$ seconds; T_s is the sampling interval; q is the multiple of frequency resolution, it is confined in the scope $q = -N'/2, -N'/2+1, \dots, N'/2$; M is the interval of the cyclic-frequency resolution; L is the overlap sample size of the received CM signal; f_1 and f_2 are the center frequencies of the filters used in the FAM, i.e., $f_1 \stackrel{\text{def}}{=} f + \alpha/2$ and $f_2 \stackrel{\text{def}}{=} f - \alpha/2$; $r_T(\lambda, f_1)$ and $r_T(\lambda, f_2)$ are the complex demodulates of $r'(n)$; $r_T^*(\lambda, f_2)$ denotes the complex conjugate of $r_T(\lambda, f_2)$. Note that

$$r_T(n, f) \stackrel{\text{def}}{=} \sum_{\lambda=-N'/2}^{N'/2-1} w(\lambda) r'(n - \lambda) e^{-j2\pi f(n - \lambda) T_s}, \quad (12)$$

where $w(\lambda)$ denotes a data-tapering window of the duration $T = N' T_s$ seconds, and its bandwidth coincides with the frequency resolution $\Delta f \stackrel{\text{def}}{=} 1/\Delta t = 1/(NT_s)$ of the cyclic spectrum. According to Eqs. (11), (12), the three-dimensional SCS $R_r^\alpha(f)$ of a CM signal can be obtained. It is obvious that

$R_r^\alpha(f) \geq 0$ for all f and for all α . The normalized SCS (NSCS) $\bar{R}_r^\alpha(f)$ can thus be obtained as

$$\bar{R}_r^\alpha(f) \stackrel{\text{def}}{=} \frac{R_{r_T}^\alpha(n, f)_{\Delta t}}{\max_{\alpha, f} R_{r_T}^\alpha(n, f)_{\Delta t}}. \quad (13)$$

Meanwhile, one can sample the NSCS $\bar{R}_r^\alpha(f)$ to obtain the “discrete NSCS (DNSCS)” $\bar{R}_r^{\alpha_u}(f_v)$, $u = 1, 2, \dots, 2N+1$ and $v = 1, 2, \dots, N'+1$ at $2N+1$ discrete cyclic frequencies α_u as given by

$$\alpha_u \stackrel{\text{def}}{=} -T_s + \frac{u-1}{NT_s}, \quad u = 1, 2, \dots, 2N+1 \quad (14)$$

and $N'+1$ discrete spectral frequencies f_v as given by

$$f_v \stackrel{\text{def}}{=} -\frac{1}{2T_s} + \frac{v-1}{N'T_s}, \quad v = 1, 2, \dots, N'+1. \quad (15)$$

B. CPP Extraction

For ten ideal (noise-free) CM signals specified in the CCSDS standard [5], the magnitudes of the corresponding DNSCSs and their grayscale colormaps with respect to f and α are illustrated by Fig. 3. It is apparent from Fig. 3 that each of such colormaps of the DNSCSs of the CM signals demonstrates four symmetric patterns on the f - α plane; the subimage of each pattern region resembles an animal's paw print. Hence, we call the RGB colormap of the magnitudes of a DNSCS a “cyclic-paw-print (CPP)”. It is conspicuous that the CPPs resulting from different CM signals are quite distinguishable as illustrated by Fig. 3.

Once the DNSCS $\bar{R}_r^{\alpha_u}(f_v)$ is obtained from a CM signal according to Eqs. (11)-(15), the magnitudes of the DNSCS $\bar{R}_r^{\alpha_u}(f_v)$ can be treated as a colormap (image) formulated as the $(2N+1) \times (N'+1)$ matrix $\tilde{R} \stackrel{\text{def}}{=} [\gamma_{u,v}]_{1 \leq u \leq 2N+1, 1 \leq v \leq N'+1}$, where

$$\gamma_{u,v} \stackrel{\text{def}}{=} |\bar{R}_r^{\alpha_u}(f_v)|, \quad u = 1, 2, \dots, 2N+1 \text{ and } v = 1, 2, \dots, N'+1. \quad (16)$$

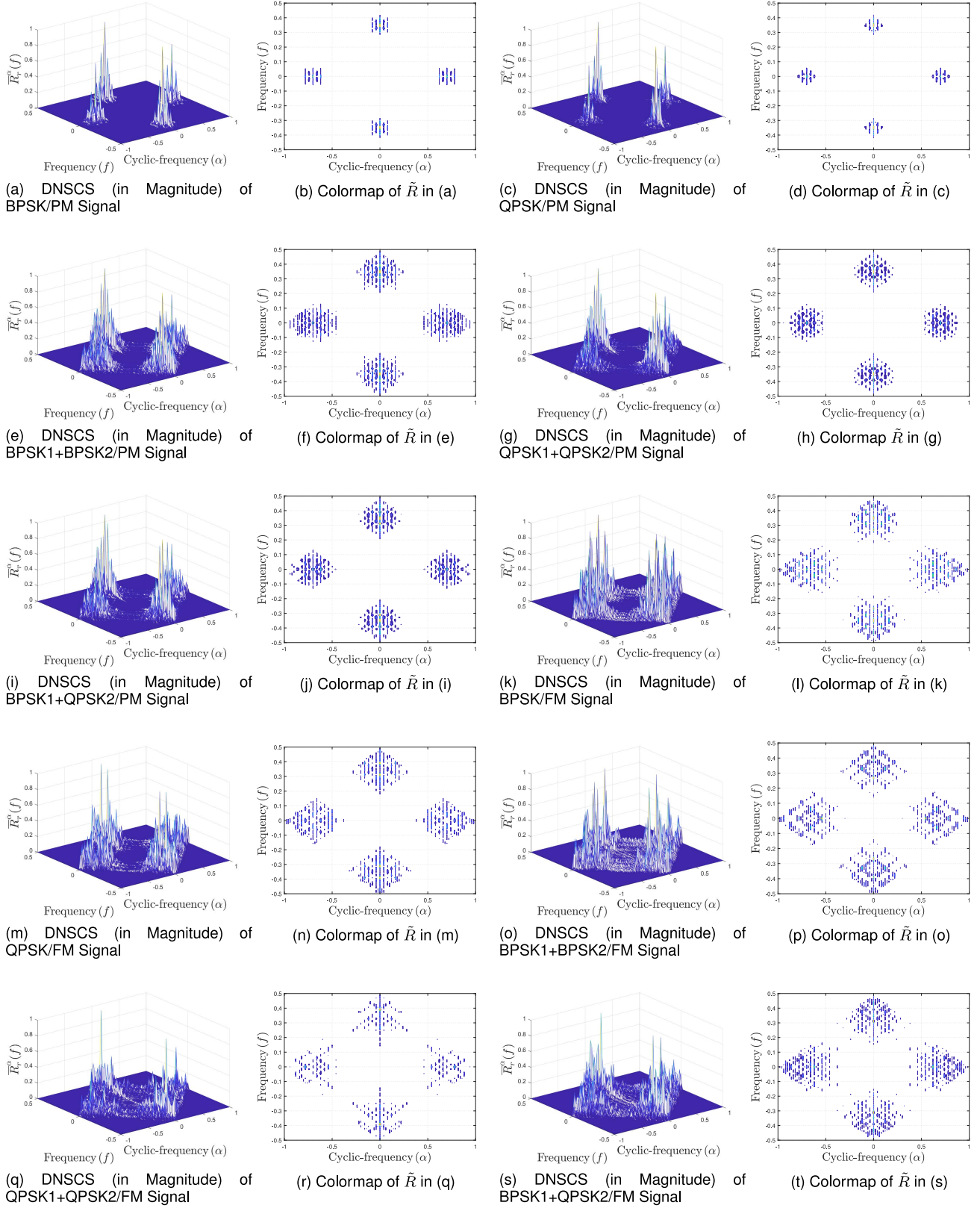


Fig. 3. Illustration of the magnitudes of the DNSCSs and the corresponding grayscale colormaps of ten CM signals defined in the CCSDS standard [5].

Then, the grayscale colormap $\tilde{R}$ is further converted to a “quantized” image matrix $\tilde{R}' \stackrel{\text{def}}{=} [\gamma'_{u,v}]_{1 \leq u \leq 2N+1, 1 \leq v \leq N'+1}$ by a b -bit quantizer such that

$$\gamma'_{u,v} \stackrel{\text{def}}{=} \lfloor 2^{b-1} \times \gamma_{u,v} \rfloor, \quad u = 1, 2, \dots, 2N+1 \text{ and } v = 1, 2, \dots, N'+1, \quad (17)$$

where b is a positive integer and $\lfloor \cdot \rfloor$ denotes the floor function which outputs the greatest integer less than or equal to the real input number. Without loss of generality, b can be set to 16 in this work. Note that we usually have $N' \ll 2N$, which results in a rectangular grayscale $\tilde{R}'$. To suit our proposed new ULWNet (to be introduced in Section III-C), the rectangular grayscale colormap $\tilde{R}'$ needs to be reshaped into a $\Lambda \times \Lambda$ square matrix $\tilde{\mathcal{R}} \stackrel{\text{def}}{=} [\gamma_{p,q}^\dagger]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda}$ by interpolating each row vector (extending $2N + 1$ entries to Λ entries for each row of $\tilde{R}'$ and decimating each column vector (reducing $N' + 1$ entries to Λ entries for each column of $\tilde{R}'$ subject to $V < \Lambda < U$). According to [42], the grayscale image matrix $\tilde{\mathcal{R}}$ can be further transformed into a pseudo-color square image, namely the “*parula colormap*” using the corresponding function in MATLAB such that

$$\gamma_{p,q}^\dagger \xrightarrow{\text{parula}} \tilde{\mathcal{R}}_{p,q}^{\text{RGB}}, \quad p = 1, 2, \dots, \Lambda \text{ and } q = 1, 2, \dots, \Lambda, \quad (18)$$

where

$$\tilde{\mathcal{R}}_{p,q}^{\text{RGB}} \stackrel{\text{def}}{=} [\mathcal{R}_{p,q}^{\text{R}}, \mathcal{R}_{p,q}^{\text{G}}, \mathcal{R}_{p,q}^{\text{B}}]^T, \quad p = 1, 2, \dots, \Lambda \text{ and } q = 1, 2, \dots, \Lambda. \quad (19)$$

According to Eqs. (18) and (19), three square matrices can be obtained for the individual redness (R), greenness (G), and blueness (B) channels such that

$$\tilde{\mathcal{R}}_{\text{R}} \stackrel{\text{def}}{=} [\mathcal{R}_{p,q}^{\text{R}}]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda}, \quad (20)$$

$$\tilde{\mathcal{R}}_{\text{G}} \stackrel{\text{def}}{=} [\mathcal{R}_{p,q}^{\text{G}}]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda}, \quad (21)$$

$$\tilde{\mathcal{R}}_{\text{B}} \stackrel{\text{def}}{=} [\mathcal{R}_{p,q}^{\text{B}}]_{1 \leq p \leq \Lambda, 1 \leq q \leq \Lambda}. \quad (22)$$

Consequently, the CPP, which is the square pseudo-color image, can be represented by the three-order tensor

$$\mathcal{R}_{\text{RGB}} \stackrel{\text{def}}{=} [\tilde{\mathcal{R}}_{\text{R}}, \tilde{\mathcal{R}}_{\text{G}}, \tilde{\mathcal{R}}_{\text{B}}]. \quad (23)$$

C. Ultra Lightweight Network (ULWNet) for Classification

Once the CPP (in terms of $\mathcal{R}_{\text{RGB}}$ given by Eq. (23)) of a received CM signal is obtained, our proposed novel ultra lightweight network (ULWNet) can be efficiently trained to perform ACMC. Our proposed ULWNet, which is based on a novel “*shallow*”¹ architecture to not only lead to a promising classification performance but also maintain a low computational complexity, consists of five convolution-layer groups (namely E-units), a max-pooling layer, a flatten layer, and a fully-connected layer. The overall architecture of ULWNet is illustrated by Fig. 2. The five convolution-layer groups (E-units) share the same structure. In the first E-unit, this convolution layer specified by sixty-four filters (kernels) of size 7×7 with the stride of two is employed to capture more features from a CPP image, and the kernel size of the convolution layers in the four subsequent E-units is reduced to 3×3 to limit the incurred computational complexity. The convolution layer in each E-unit is followed by a batch

normalization (BN) layer and an exponential linear unit (eLU) activation layer. The CPP of the received CM signal, $\mathcal{R}_{\text{RGB}}$, which is represented as a tensor composed by three channels, is directly applied to the first convolution layer within the E-Unit. The input of the convolution layer in the ℓ^{th} E-unit, $\ell = 1, 2, \dots, 5$ is denoted by a $\Upsilon_\ell^{\text{th}}$ -order tensor:

$$\mathbf{Z}_\ell \stackrel{\text{def}}{=} [\tilde{Z}_{1,\ell}, \tilde{Z}_{2,\ell}, \dots, \tilde{Z}_{\Upsilon_\ell,\ell}], \quad (24)$$

where $\tilde{Z}_{v,\ell}$ is the v^{th} feature map in the ℓ^{th} E-unit, $v = 1, 2, \dots, \Upsilon_\ell$. Here, for the convolution layer in the first E-unit ($\ell=1$), Υ_1 is fixed to 3 while each other Υ_ℓ , $\ell = 2, 3, \dots, 5$ is equal to the number of kernel filters of the previous convolution layer $\mathcal{N}_{\ell-1}$. The kernel filters of the convolution layer in the ℓ^{th} E-unit are denoted by $\mathbf{W}_\ell \stackrel{\text{def}}{=} [\tilde{W}_{1,\ell}, \tilde{W}_{2,\ell}, \dots, \tilde{W}_{\mathcal{N}_\ell,\ell}]$ where $\tilde{W}_{n,\ell}$ expresses the n^{th} two-dimensional convolution filter in the ℓ^{th} E-unit, $n = 1, 2, \dots, \mathcal{N}_\ell$. The output of the convolution layer in the ℓ^{th} E-unit can be calculated as

$$\hat{\mathbf{Z}}_\ell \stackrel{\text{def}}{=} \mathbf{W}_\ell \otimes \mathbf{Z}_\ell + \mathbf{B}_\ell, \quad (25)$$

where “ $\otimes$ ” denotes the convolution between two tensors, $\mathbf{B}_\ell \stackrel{\text{def}}{=} [\tilde{B}_{1,\ell}, \tilde{B}_{2,\ell}, \dots, \tilde{B}_{\mathcal{N}_\ell,\ell}]$ is the bias for the convolution layer in the ℓ^{th} E-unit. Note that $\hat{\mathbf{Z}}_\ell$ is a tensor composed of $\mathcal{N}_\ell \times \mathcal{M}_\ell$ square matrices, and the size of each involved square matrix, namely $\mathcal{M}_\ell$, for each channel in our framework can be updated by

$$\mathcal{M}_\ell \stackrel{\text{def}}{=} \left\lfloor \frac{\mathcal{M}_\ell + 2\mathcal{P}_\ell - \mathcal{K}}{\mathcal{S}} \right\rfloor + 1, \quad (26)$$

where $\mathcal{M}_\ell$ is the size of $\tilde{Z}_{v,\ell}$ in $\mathbf{Z}_\ell$ (see Eq. (24)) for the input of the convolution layer in the ℓ^{th} E-unit; $\mathcal{P}_\ell$, $\mathcal{K}$, and $\mathcal{S}$ denote the padding size, the kernel size, and the stride size, respectively. In this work, we set $\mathcal{K} = 3$ and $\mathcal{S} = 2$. Moreover, we set

$$\mathcal{P}_\ell = \begin{cases} 3, & \text{if } \ell = 1 \\ 1, & \text{if } \ell = 2, 3, 4, 5. \end{cases} \quad (27)$$

Next, $\hat{\mathbf{Z}}_\ell$ is normalized subject to its mean μ_ℓ and variance σ_ℓ^2 in the following BN layer. The output of the BN layer is thus given by

$$\bar{\mathbf{Z}}_\ell \stackrel{\text{def}}{=} \frac{\varrho_\ell}{\sqrt{\sigma_\ell^2 - \varepsilon}} (\hat{\mathbf{Z}}_\ell - \mu_\ell \mathbf{I}) + \xi_\ell \mathbf{I}, \quad (28)$$

where $\mathbf{I}$ denotes a tensor with all 1 elements and same dimensions as $\hat{\mathbf{Z}}_\ell$; ε is a constant to ensure the numerical stability and we usually set $\varepsilon = 10^{-5}$; the scale factor ϱ_ℓ and the offset parameter ξ_ℓ are updated during training. Finally, a nonlinear activation function eLU defined in [43] is applied to $\bar{\mathbf{Z}}_\ell$ to accelerate the convergence and improve the classification accuracy. The output of eLU in the ℓ^{th} E-unit, denoted by $\check{\mathbf{Z}}_\ell$, is employed as the input of the $(\ell+1)^{\text{th}}$ E-unit, i.e., $\mathbf{Z}_{\ell+1} = \check{\mathbf{Z}}_\ell$ and $\mathcal{M}_{\ell+1} = \mathcal{M}_\ell$, $\ell = 1, 2, \dots, 5$.

In our proposed ULWNet, the output of the last E-unit, $\mathbf{Z}_6$, is further processed by a max-pooling layer, a flatten layer, and a fully-connected layer in series to result in a K -dimensional feature vector $\vec{D} \stackrel{\text{def}}{=} [d_1, d_2, \dots, d_K]^T$ where K corresponds to

¹When a DL model has a *shallow* architecture, it implies that there involve only a few convolution layers.

TABLE I
PARAMETERS OF CM SIGNALS

Parameter	Specification
Radio Frequency	8GHz
Intermediate Frequency (IF)	70MHz
Subcarrier 1's Frequency	8kHz
Subcarrier 2's Frequency	16kHz
Symbol Rate	4kSps
Sampling Rate	200MHz
Roll-off Factor	0.35
Transmission Channel	AWGN
Signal-to-Noise-Ratio (SNR)	-20dB, 20dB

the number of CM (candidate) schemes under consideration and $\sum_{k=1}^K d_k = 1$. Therefore, a softmax activation function as defined in [44] can be utilized to identify the CM scheme of a received signal such that

$$\varsigma_k = \text{softmax}(d_k), \quad k = 1, 2, \dots, K, \quad (29)$$

where ς_k corresponds to the estimated *a priori* probability of the k^{th} CM scheme $\mathcal{K}_k, k=1, 2, \dots, K$ in the candidate set $\mathbb{K} \stackrel{\text{def}}{=} \{\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_K\}$ and $\varsigma \stackrel{\text{def}}{=} [\varsigma_1, \varsigma_2, \dots, \varsigma_K]^T$. The CM scheme of the received signal is ultimately identified as $\mathcal{K}_{\hat{k}}$ where

$$\hat{k} \stackrel{\text{def}}{=} \underset{k \in \{1, 2, \dots, K\}}{\text{argmax}} \quad \varsigma_k. \quad (30)$$

IV. SIMULATION

In Section IV-A, the effectiveness of our proposed new ACMC approach is first evaluated via Monte Carlo simulations in terms of *probability of correct classification* (P_{cc}) for each CM scheme and the *correct recognition percentage* (CRP) over the entire CM candidate set $\mathbb{K}$ versus the SNR η in dB. Then, the robustness of our proposed new ACMC approach is also tested in Section IV-B. Finally, the computational complexity of our proposed new ACMC scheme is evaluated in Section IV-C.

Here, the CM candidate set $\mathbb{K}$ adopted for our simulations includes ten CM schemes specified in [5], namely

1. PCM/BPSK/PM,
2. PCM/QPSK/PM,
3. PCM/(BPSK1+BPSK2)/PM,
4. PCM/(QPSK1+QPSK2)/PM,
5. PCM/(BPSK1+QPSK2)/PM,
6. PCM/(BPSK1+BPSK2)/FM,
7. PCM/(QPSK1+QPSK2)/FM,
8. PCM/(BPSK1+QPSK2)/FM,
9. PCM/BPSK/FM,
10. PCM/QPSK/FM,

where the list numbers serve as the CM scheme indices. The corresponding CM signals are generated according to the parameters defined by Table I. According to this CM candidate set $\mathbb{K}$, BPSK, QPSK, or the combination of both is utilized for the inner-layer modulation of the single-subcarrier CM signals while PM or FM is adopted for the outer-layer modulation. These CM signals are transmitted over a wireless channel and acquired by an intelligent receiver, which identifies the CM scheme of the received signal using our

proposed new ACMC method. The modulation-indices K_p and K_f for the primary modulation (PM or FM) are both set to 1.2, the FFT window size for generating the SCS in the FAM is fixed to $N' = 64$, and the sample size is set to $N=1024$. Note that $\tilde{R}$ is a 2049×65 matrix, which is further transformed into a $224 \times 224 \times 3$ pseudo-color image tensor. In this work, the AWGN channel, the Rayleigh channel, and the α -stable distributed noise channel are all considered in our simulations, and the corresponding training/testing datasets are also constructed respectively. The SNRs for the AWGN channel and the Rayleigh channel and the mixed SNR (MSNR) for the α -stable distributed noise channel all range from -20 to 20dB with an interval of 5dB. For the signal modulated by the k^{th} CM-scheme $\mathcal{K}_k \in \mathbb{K}, k=1, 2, \dots, K$ at each SNR or MSNR in the aforementioned range, five hundred Monte Carlo trials are taken. In each trial, 2048 signal samples are randomly generated by computer and employed to calculate a CPP image using our proposed new ACMC approach. Consequently, five hundred CPP images are produced in total for the k^{th} CM-scheme $\mathcal{K}_k \in \mathbb{K}$ at each SNR or MSNR in the above-stated range. Now we have ten different candidate CM-schemes, i.e., $K=10$. Therefore, the training and testing datasets at each SNR or MSNR in the aforementioned range consists of five thousand CPP images totally. Then, such CPP images are further divided into the training and testing sets with respect to a ratio of 7:3, where the training set consists of three thousand and five hundred CPP images and the testing set contains one thousand and five hundred CPP images for each SNR or MSNR condition.

Note that we set $\mathcal{J} = 1$ (only one sub-carrier) for the PCM/BPSK/PM, PCM/QPSK/PM, PCM/BPSK/FM, and PCM/QPSK/FM while we set $\mathcal{J} = 2$ (two subcarriers) for PCM/(BPSK1+BPSK2)/PM, PCM/(QPSK1+QPSK2)/PM, PCM/(BPSK1+QPSK2)/PM, PCM/(BPSK1+BPSK2)/FM, PCM/(QPSK1+QPSK2)/FM, and PCM/(BPSK1+QPSK2)/FM. Besides, we randomly generate the information symbols $a_{i,q}$'s by computer where $a_{i,q}$'s are drawn from a binary uniform random process and $a_{i,q} \in \{0, 1\}$ for all i, q . Additionally, $T_1=0.05\text{msec}$, $T_2=0.04\text{msec}$, and $T_s=0.05\mu\text{sec}$. The received CM signals are sampled accordingly to acquire the corresponding CPPs. The CPP feature tensors given by Eq. (23) are then randomly divided into the training and validation subsets subject to the ratio of 7:3. Adam optimizer is employed during training while the learning rate for updating each parameter is fixed to 0.0001. The mini-batch of a size 64 is also adopted to accelerate the convergence and improve the computational efficiency. The maximum number of training epochs is set to be 200. The hyperparameters of the ULWNet and the corresponding values are presented in Table II below.

A. P_{cc} and CRP Evaluation of Our Proposed New ACMC Approach

The CRPs of our proposed new ACMC approach over all candidates in $\mathbb{K}$ are evaluated subject to different SNRs, which are delineated by Fig. 4. According to Fig. 4, the CRP resulting from our proposed new ACMC approach can lead to the perfect performance (100%) when $\eta \geq 5\text{dB}$. The

TABLE II
THE HYPERPARAMETERS AND THE CORRESPONDING
VALUES OF OUR PROPOSED ULWNet

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.0001
Mini-batch	64
Training Epochs	200

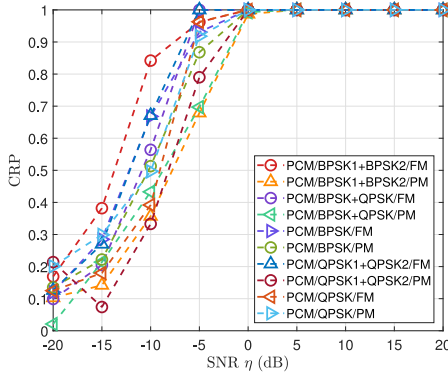


Fig. 4. CRPs versus SNR η for ten CM schemes in $\mathbb{K}$ resulting from our proposed new ACMC approach where an AWGN channel is considered.

confusion matrices for the aforementioned ten CM schemes in $\mathbb{K}$ resulting from our proposed new ACMC approach for $\eta = -10$ dB, 0 dB, and 10 dB are presented by Fig. 5. According to Figure 5, when the SNR is equal to 0 dB, our proposed new ACMC approach can perfectly recognize the CM schemes in $\mathbb{K}$ except PCM/QPSK/PM and PCM/(QPSK1+QPSK2)/PM while the CRPs for these two CM schemes resulting from our proposed new approach can still reach up to 99%.

Meanwhile, the probabilities of correct classification P_{cc} 's for the aforementioned CM-scheme set $\mathbb{K}$ resulting from our proposed new ACMC approach and the existing HOS-based ACMC scheme proposed in [20] by considering an AWGN channel are delineated by Fig. 6. It is well known that the HOS approach is quite sensitive to the sample size. Therefore, we investigate several different sample sizes, namely $N=2048$, 8192, 16384, 24567, 32768, and 40960. According to Fig. 6, our proposed new ACMC approach can produce the perfect classification performance ($P_{cc} = 100\%$) even when the smallest sample size ($N = 2048$) is adopted and $\eta \geq 0$ dB. However, the existing HOS-based ACMC method cannot lead to any satisfactory classification performance ($P_{cc} \leq 70\%$ over the entire SNR regime) no matter how many samples are adopted.

Furthermore, to evaluate the effectiveness of our proposed CPP features, other features, including I/Q features in [29], cyclostationary features in [24], and FBREWT features in [25], are also extracted for the ten CM schemes in $\mathbb{K}$ and employed to train our proposed ULWNet. The P_{cc} 's for the aforementioned candidate CM-scheme set $\mathbb{K}$ in the presence of an AWGN channel resulting from our proposed new ACMC approach using CPP, I/Q, cyclostationary, and FBREWT features are delineated by Fig. 7.

According to Fig. 7, the P_{cc} curve resulting from our proposed new ACMC method using CPP features is always above those resulting from our proposed new ACMC method

using I/Q, cyclostationary, and FBREWT features. Thus, our proposed new CPP features are the most robust features for the DL network.

B. Robustness of Our Proposed New ACMC Approach

1) *The Robustness of Our Proposed New ACMC Technique Against Various Channels:* The robustness of our proposed new ACMC approach is first investigated by evaluating its performances in the presence of AWGN channel, α -stable distributed noise channel (see [45]), and Rayleigh channel. Other simulation parameters remain the same as presented before. For the aforementioned CM-scheme set $\mathbb{K}$, the P_{cc} 's resulting from our proposed new ACMC approach in the presence of AWGN and Rayleigh channels are depicted by Fig. 8. The P_{cc} 's resulting from our proposed new ACMC approach (denoted by "ULWNet-Rayleigh" in Fig. 8) are significantly degraded under the Rayleigh channel in low SNR conditions in comparison with the P_{cc} 's under the AWGN channel (denoted by "ULWNet-AWGN" in Fig. 8). However, the P_{cc} 's resulting from our proposed new ACMC approach can still reach to 100% under the Rayleigh channel when $\eta \geq 10$ dB.

On the other hand, we would like to investigate our proposed new ACMC scheme under the α -stable distributed noise channel stated in [45] as well. The "mixed signal-to-noise ratio (MSNR)" is often employed to quantify the condition of an α -stable distributed noise channel according to [45]. Here, α is set to be 1.5, and other parameters stay the same as presented before. The results are plotted by Fig. 9. By comparing Figs. 8 and 9, one can find that our proposed new ACMC approach can lead to almost the same classification performance for all SNRs/MSNRs in the presence of AWGN channel and α -stable distributed noise channel. Thus, our proposed new ACMC approach is robust against both AWGN channel and α -stable distributed noise channel but slightly sensitive to the Rayleigh channel.

2) *The Robustness of Our Proposed CPP Features Against Different DL Networks:* Then, the superiority of our proposed new deep-learning model, namely ULWNet, to three other existing deep-learning models: MobileNetV2 in [46], ResNet in [47], and ConvNeXt in [48] is justified here. All the deep-learning models in comparison adopt the CPP features while the corresponding P_{cc} 's are evaluated. The CM-scheme set $\mathbb{K}$ and the simulation parameters remain unchanged as presented before. The P_{cc} 's resulting from the aforementioned four deep-learning models are demonstrated by Fig. 10.

According to Fig. 10, all four deep-learning models in comparison lead to almost the same classification performance when the input is fed by the CPP features. As a result, our proposed CPP features are very robust and insensitive to the choice of any deep-learning model (as its classifier).

3) *The Robustness of Our Proposed New ACMC Technique Against Various Residual Synchronization Errors:* The robustness of our proposed new ACMC technique is also investigated to evaluate the corresponding performances in the presence of timing errors, phase offsets, and frequency offsets. Assume

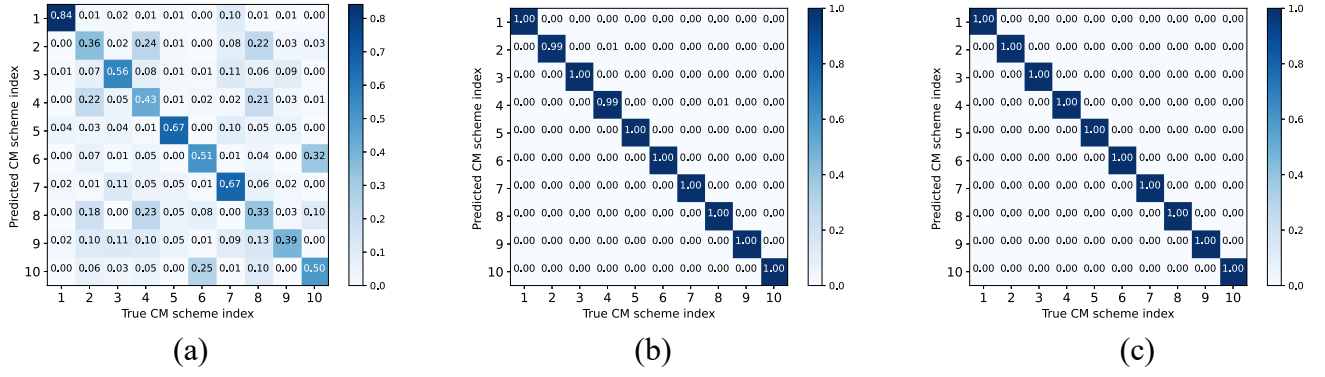


Fig. 5. Confusion matrices for ten CM schemes in $\mathbb{K}$ where an AWGN channel is considered.

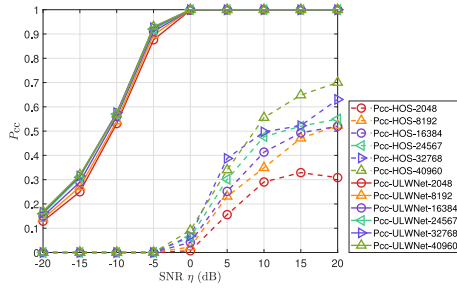


Fig. 6. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ resulting from our proposed new ACMC approach and the existing HOS-based ACMC scheme proposed in [20] where an AWGN channel is considered. Several sample sizes, namely $N=2048, 8192, 16384, 24567, 32768$, and 40960 are investigated here.

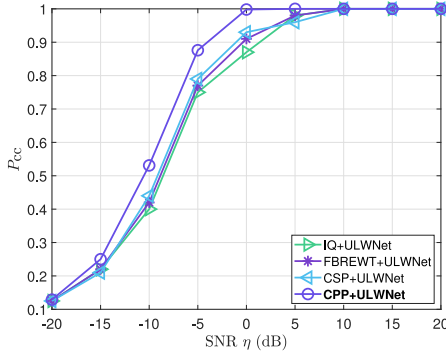


Fig. 7. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ resulting from our proposed new ACMC approach using CPP, I/Q, cyclostationary, and FBREWT features where an AWGN channel is considered. The sample size is $N=2048$.

that the received CM signal suffers from the residual timing error due to imperfect timing-synchronization at the CM receiver. Usually, the maximum timing-error which a typical timing synchronizer can handle is approximately 10% of a symbol period T_s according to [49]. Thus, the residual timing-error here is set to be 2.5%, 5%, 7.5%, and 10% of T_s , respectively. Other simulation settings remain unchanged as before. The P_{cc} 's for the ten CM-scheme candidates in $\mathbb{K}$ resulting from our propose new ACMC method subject to the aforementioned timing errors are evaluated and compared with those resulting from our proposed new ACMC method subject to perfect timing synchronization in Fig. 11. It is conspicuous

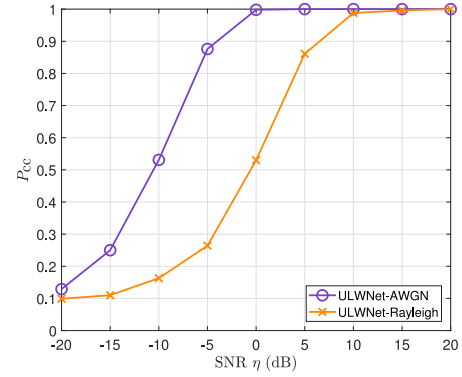


Fig. 8. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ resulting from our proposed new ACMC approach where an AWGN channel and a Rayleigh channel are considered.

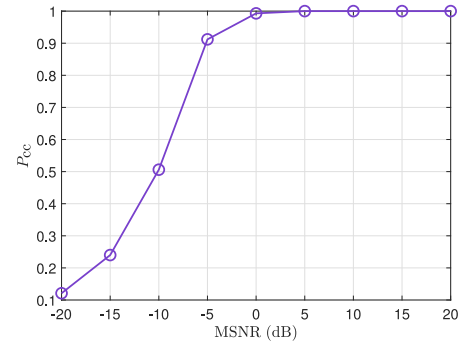


Fig. 9. Probabilities of correct classification versus MSNR for ten CM schemes in $\mathbb{K}$ resulting from our proposed new ACMC approach where an α -stable distributed noise channel is considered.

from Fig. 11 that our proposed new ACMC technique almost leads to the same P_{cc} 's for all different types of timing errors in the SNR regime. As a result, our proposed new ACMC approach is very robust against timing errors.

Next, the impact of carrier-frequency offset on the performance of our proposed new ACMC technique is investigated. According to [50], [51], the bandwidth of the CM signal utilized in the typical satellite telemetry transmission system is 10MHz and its initial frequency error is up to 2MHz. To emulate the intermediate frequency (IF) sampling in a real CM receiver, the primary carrier frequency f_c is set to be the IF 70MHz while the residual frequency-offset is set to be $0.025f_c$,

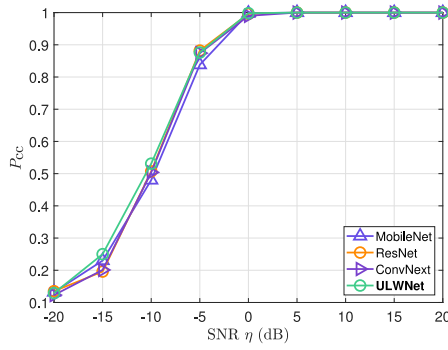


Fig. 10. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ resulting from our proposed new deep-learning model (ULWNet) and three other existing deep-learning models: MobileNetV2 in [46], ResNet in [47], and ConvNeXt in [48] where an AWGN channel is considered.

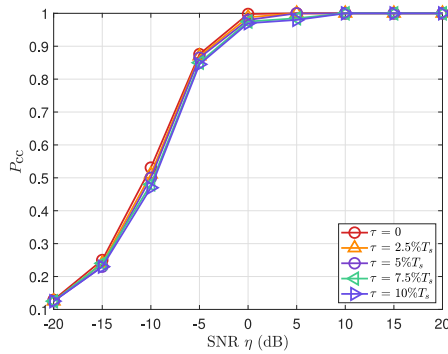


Fig. 11. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ subject to different residual timing-errors resulting from our proposed new ACMC approach where an AWGN channel is considered. The sample size is $N=2048$.

$0.05f_c$, $0.075f_c$, and $0.1f_c$, respectively. Other simulation settings remain the same as before. The P_{cc} 's for the ten CM-scheme candidates in $\mathbb{K}$ resulting from our proposed new ACMC method subject to the aforementioned frequency-offsets are evaluated and compared with those resulting from our proposed new ACMC method subject to perfect frequency synchronization in Fig. 12. According to Fig. 12, the P_{cc} 's resulting from our proposed new ACMC method subject to non-zero frequency-offsets appear to be slightly worse than those subject to perfect frequency-synchronization (zero frequency-offset) when the SNR ranges from -15dB to 10dB . Nonetheless, the P_{cc} 's subject to non-zero frequency offsets all can still reach up to 100% when $\text{SNR} \geq 10\text{dB}$. Thus, our proposed new ACMC approach can lead to a fairly consistent performance against various kinds of frequency-offsets.

At last, the effect of carrier-phase offset on the performance of our proposed new ACMC technique is investigated here. Suppose that the phase offset cannot be completely removed by the carrier synchronization loop at the CM receiver. Since the linear response interval of the phase detector in the carrier synchronization loop adopted by the CM receiver ranges from $-\pi/4$ to $\pi/4$ according to [52], the residual phase offset is set to be $\pi/16$, $\pi/8$, $3\pi/16$, and $\pi/4$ (the worst case), respectively. Other simulation settings remain the same as before. The P_{cc} 's for the ten CM-scheme candidates in $\mathbb{K}$ resulting from

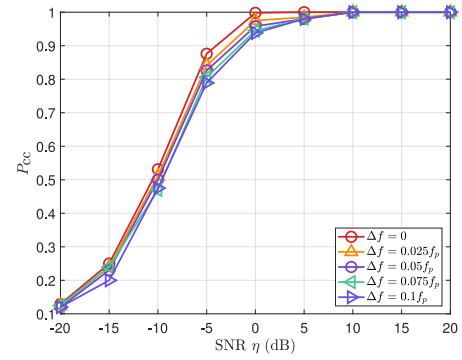


Fig. 12. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ subject to different residual frequency-offsets resulting from our proposed new ACMC approach where an AWGN channel is considered. The sample size is $N=2048$.

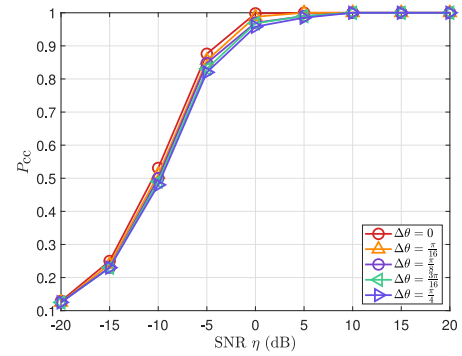


Fig. 13. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ subject to different residual phase offsets resulting from our proposed new ACMC approach where an AWGN channel is considered. The sample size is $N = 2048$.

our proposed new ACMC method subject to the above-stated phase offsets are evaluated and compared with those resulting from our proposed new ACMC method subject to perfect phase synchronization (zero phase offset) in Fig. 13. It is obvious from Fig. 13 that the performance of our proposed new ACMC method slightly deteriorates in the presence of non-zero phase offset when $-15\text{dB} \leq \text{SNR} \leq 10\text{dB}$. The P_{cc} 's subject to non-zero phase offsets can still reach up to 100% when $\text{SNR} \geq 10\text{dB}$. Therefore, our proposed new ACMC method is quite robust against phase offsets.

C. Computational-Complexity Analysis

Since the classification network in a feature-based ACMC method contributes a major part of the overall computational complexity, the effect of the number of E-units involved in our proposed new ULWNet on the classification performance should also be investigated here. Eight different ULWNet architectures, which include two, three, four, five, six, seven, eight, and nine E-units totally, in our proposed new ACMC approach are constructed respectively. The corresponding P_{cc} 's for the aforementioned CM-scheme set $\mathbb{K}$ in the presence of an AWGN channel are depicted by Fig. 14. It's obvious that our proposed new ACMC approach can result in almost the same classification performance for all SNRs when the ULWNet contains five or more E-units. In addition, the P_{cc} 's

TABLE III
THE COMPUTATIONAL COMPLEXITIES OF OUR PROPOSED NEW ACMC SCHEME AND THE EXISTING HOS- AND DL-BASED ACMC METHODS

	Mobilenet V2	ResNet	ConvNet	HOS	ULWNet
Computational Complexity (FLOPs)	319.03×10^9	259.36×10^9	15.38×10^9	-	0.51×10^9
Number of Trainable Parameters	2.24×10^6	2.35×10^6	8.76×10^6	-	5.48×10^5
Training Time	2.72 hours	2.88 hours	4.94 hours	-	2.28 hours
Testing Time	529 msec	529.67 msec	531 msec	108 msec	527.67 msec

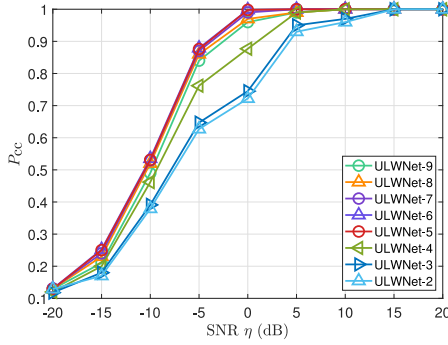


Fig. 14. Probabilities of correct classification versus SNR η for ten CM schemes in $\mathbb{K}$ resulting from our proposed new ACMC approach involving various ULWNet architectures, which have 2, 3, ..., 9 E-units, respectively, where an AWGN channel is considered.

produced by our proposed new ACMC approach slightly degrade when the ULWNet involves more than seven E-units. Consequently, there are five E-units in the ULWNet of our proposed new ACMC scheme.

Since the computational complexities of the DL-based ACMC schemes cannot be analytically expressed by $\mathcal{O}$ -functions, the computational complexity of our proposed new ACMC scheme can only be estimated in terms of the required *floating-point operations* (FLOPs), the size of trainable parameters, the training and testing time lapses. The computational complexities of the existing HOS-based ACMC method in [20] and the CPP-based ACMC methods using different DL-classification networks (MobileNetV2 in [46], ResNet in [47], and ConvNeXt in [48]) are also evaluated for comparison. In our simulations, all ACMC methods are undertaken using the workstation configured with an Intel Xeon GOLD 5222 CPU, 64 GB RAM, and a double NVIDIA GeForce RTX 2080TI GPU. The computational-resources and run-times of our proposed new ACMC scheme, the existing HOS-based ACMC scheme, and three other DL-based ACMC schemes are summarized by Table III. According to Table III, the existing HOS-based ACMC method takes the least time due to its simple features. However, according to Fig. 6, the classification performance of the HOS-based ACMC method is unsatisfactory. As for the DL-based ACMC methods, according to Table III, our proposed ULWNet requires less computational-resource and run-time than the three other DL-based ACMC schemes (MobileNetV2, ResNet, and ConvNeXt).

V. CONCLUSION

In this work, we introduce a novel automatic composite-modulation classification approach using the CPP features and

our proposed new ultra lightweight network (ULWNet) to identify the modulation scheme of a received CM signal in cognitive intelligent telemetry, tracking, and command as well as broadband space communications. First, the CPP features are built upon the second-order cyclic spectrum of the received CM signal. Then, the resource-efficient ULWNet is trained to classify the CM scheme using the extracted CPP features as the input. Monte Carlo simulation results justify the superiority of our proposed new ACMC approach to the existing ACMC methods. Our proposed new robust ACMC technique would be a promising solution to the next generation intelligent telemetry, tracking, and command systems for cognitive space communications and space surveillance.

REFERENCES

- [1] C. Wu, K.-L. A. Yau, Z. Zhang, D. Turgut, and S. Mao, "Guest editorial special issue on collaborative intelligence for green Internet of Things in the 6G era," *IEEE Trans. Green Commun. Netw.*, vol. 7, no. 2, pp. 855–856, Jun. 2023.
- [2] H. Tataria, M. Shafi, A. F. Molisch, M. Dohler, H. Sjöland, and F. Tufvesson, "6G wireless systems: Vision, requirements, challenges, insights, and opportunities," *Proc. IEEE*, vol. 109, no. 7, pp. 1166–1199, Jul. 2021.
- [3] H. Jiang, Y. Niu, B. Ai, Z. Zhong, and S. Mao, "QoS-aware bandwidth allocation and concurrent scheduling for terahertz wireless backhaul networks," *IEEE Access*, vol. 8, pp. 125814–125825, 2020.
- [4] P. Ramezani, B. Lyu, and A. Jamalipour, "Toward RIS-enhanced integrated terrestrial/non-terrestrial connectivity in 6G," *IEEE Netw.*, vol. 37, no. 3, pp. 178–185, May/Jun. 2023.
- [5] *Radio Frequency and Modulation Systems-Part 1 Earth Stations and Spacecraft*, Consultative Committee Space Data Syst. (CCSDS), Washington, DC, USA, Feb. 2020.
- [6] J. Vilà-Valls, M. Navarro, P. Closas, and M. Bertinelli, "Synchronization challenges in deep space communications," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 34, no. 1, pp. 16–27, Jan. 2019.
- [7] H. Xie, Y. Zhan, G. Zeng, and X. Pan, "LEO mega-constellations for 6G global coverage: Challenges and opportunities," *IEEE Access*, vol. 9, pp. 164223–164244, 2021.
- [8] C. Zhang, C. Jiang, L. Kuang, J. Jin, Y. He, and Z. Han, "Spatial spectrum sharing for satellite and terrestrial communication networks," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 3, pp. 1075–1089, Jun. 2019.
- [9] L. Lu, W. Xu, Y. Wang, and Z. Tian, "Compressive spectrum sensing using sampling-controlled block orthogonal matching pursuit," *IEEE Trans. Commun.*, vol. 71, no. 2, pp. 1096–1111, Feb. 2023.
- [10] W. Zhao, S. Ali, M. Jin, G. Cui, N. Zhao, and S. Yoo, "Extreme eigenvalues-based detectors for spectrum sensing in cognitive radio networks," *IEEE Trans. Commun.*, vol. 70, no. 1, pp. 538–551, Jan. 2022.
- [11] J. Yuan, Y. Liang, J. Joung, G. Feng, and E. G. Larsson, "Intelligent reflecting surface-assisted cognitive radio system," *IEEE Trans. Commun.*, vol. 69, no. 1, pp. 675–687, Jan. 2021.
- [12] M. Abdelbar, W. H. Tranter, and T. Bose, "Cooperative cumulants-based modulation classification in distributed networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 4, no. 3, pp. 446–461, Sep. 2018.
- [13] S. Huang, Y. Yao, Z. Wei, Z. Feng, and P. Zhang, "Automatic modulation classification of overlapped sources using multiple cumulants," *IEEE Trans. Veh. Technol.*, vol. 66, no. 7, pp. 6089–6101, Jul. 2017.
- [14] Y. Yuan, P. Zhao, B. Wang, and B. Wu, "Hybrid maximum likelihood modulation classification for continuous phase modulations," *IEEE Commun. Lett.*, vol. 20, no. 3, pp. 450–453, Mar. 2016.

- [15] Y. A. Eldemerdash, O. A. Dobre, O. Üreten, and T. Yensen, "A robust modulation classification method for PSK signals using random graphs," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 2, pp. 642–644, Feb. 2019.
- [16] S. Lin, Y. Zeng, and Y. Gong, "Modulation recognition using signal enhancement and multistage attention mechanism," *IEEE Trans. Wireless Commun.*, vol. 21, no. 11, pp. 9921–9935, Nov. 2022.
- [17] J. Li et al., "Composite multiple-mode orthogonal frequency division multiplexing with index modulation," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 3748–3761, Jun. 2023.
- [18] J. Tian, Y. Pei, Y.-D. Huang, and Y.-C. Liang, "Modulation-constrained clustering approach to blind modulation classification for MIMO systems," *IEEE Trans. Cogn. Commun. Netw.*, vol. 4, no. 4, pp. 894–907, Dec. 2018.
- [19] T. R. Kishore and K. D. Rao, "Automatic intrapulse modulation classification of advanced LPI radar waveforms," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 2, pp. 901–914, Apr. 2017.
- [20] Q. Xu, J. Wan, L. He, and Q. Wang, "Blind recognition for hybrid modulation signals based on high-order cumulant and squared spectrum feature," *Acta Electronica Sinica*, vol. 50, no. 2, pp. 426–431, Feb. 2022.
- [21] P. Chu, L. Xie, C. Dai, and Y. Chen, "Automatic modulation recognition for secondary modulated signals," *IEEE Wireless Commun. Lett.*, vol. 10, no. 5, pp. 962–965, May 2021.
- [22] L. Wang, Y. Han, P. Zhou, H. Ge, and L. Shen, "Modulation recognition of composite modulation signal based on two-fold digital receiver and goodness of fit test," in *Proc. IEEE Int. Conf. Electron. Technol. Commun. Inf. (ICETCI)*, Aug. 2021, pp. 433–437.
- [23] S. Zheng et al., "Toward next-generation signal intelligence: A hybrid knowledge and data-driven deep learning framework for radio signal classification," *IEEE Trans. Cogn. Commun. Netw.*, vol. 9, no. 3, pp. 564–579, Jun. 2023.
- [24] L. Yin, X. Xiang, and Y. Liang, "Cyclostationary feature based modulation classification with convolutional neural network in multipath fading channels," *IEEE Access*, vol. 11, pp. 105455–105465, 2023.
- [25] R. R. Yakkati, R. R. Yakkati, R. K. Tripathy, and L. R. Cenkeramaddi, "Radio frequency spectrum sensing by automatic modulation classification in cognitive radio system using multiscale deep CNN," *IEEE Sensors J.*, vol. 22, no. 1, pp. 926–938, Jan. 2022.
- [26] S. Ramjee, S. Ju, D. Yang, X. Liu, A. E. Gamal, and Y. C. Eldar, "Ensemble wrapper subsampling for deep modulation classification," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 4, pp. 1156–1170, Dec. 2021.
- [27] P. Qi, X. Zhou, S. Zheng, and Z. Li, "Automatic modulation classification based on deep residual networks with multimodal information," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 1, pp. 21–33, Mar. 2021.
- [28] E. Perenda, S. Rajendran, G. Bovet, S. Pollin, and M. Zheleva, "Evolutionary optimization of residual neural network architectures for modulation classification," *IEEE Trans. Cogn. Commun. Netw.*, vol. 8, no. 2, pp. 542–556, Jun. 2022.
- [29] S. Rajendran, W. Meert, D. Giustiniano, V. Lenders, and S. Pollin, "Deep learning models for wireless signal classification with distributed low-cost spectrum sensors," *IEEE Trans. Cogn. Commun. Netw.*, vol. 4, no. 3, pp. 433–445, Sep. 2018.
- [30] Y. Lin, Y. Tu, and Z. Dou, "An improved neural network pruning technology for automatic modulation classification in edge devices," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 5703–5706, May 2020.
- [31] H. Zhang, F. Zhou, Q. Wu, W. Wu, and R. Q. Hu, "A novel automatic modulation classification scheme based on multi-scale networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 8, no. 1, pp. 97–110, Mar. 2022.
- [32] X. Fu et al., "Lightweight automatic modulation classification based on Decentralized learning," *IEEE Trans. Cogn. Commun. Netw.*, vol. 8, no. 1, pp. 57–70, Mar. 2022.
- [33] Y. Wang, J. Yang, M. Liu, and G. Gui, "LightAMC: Lightweight automatic modulation classification via deep learning and compressive sensing," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 3491–3495, Mar. 2020.
- [34] C.-F. Teng, C.-Y. Chou, C.-H. Chen, and A.-Y. Wu, "Accumulated polar feature-based deep learning for efficient and lightweight automatic modulation classification with channel compensation mechanism," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 15472–15485, Dec. 2020.
- [35] X. Yan, G. Feng, H.-C. Wu, W. Xiang, and Q. Wang, "Innovative robust modulation classification using graph-based cyclic-spectrum analysis," *IEEE Commun. Lett.*, vol. 21, no. 1, pp. 16–19, Jan. 2017.
- [36] X. Yan, G. Liu, H.-C. Wu, and G. Feng, "New automatic modulation classifier using cyclic-spectrum graphs with optimal training features," *IEEE Commun. Lett.*, vol. 22, no. 6, pp. 1204–1207, Jun. 2018.
- [37] X. Yan, G. Liu, H.-C. Wu, G. Zhang, Q. Wang, and Y. Wu, "Robust modulation classification over α -stable noise using graph-based fractional lower-order cyclic spectrum analysis," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2836–2849, Mar. 2020.
- [38] H. Zhang, D. Le Ruyet, and M. Terré, "Spectral correlation of multicarrier modulated signals and its application for signal detection," *EURASIP J. Adv. Signal Process.*, vol. 2010, Dec. 2009, Art. no. 794246.
- [39] J. Ma and T. Qiu, "Automatic modulation classification using cyclic correntropy spectrum in impulsive noise," *IEEE Wireless Commun. Lett.*, vol. 8, no. 2, pp. 440–443, Apr. 2019.
- [40] T. Liu, T. Qiu, and S. Luan, "Cyclic frequency estimation by compressed cyclic correntropy spectrum in impulsive noise," *IEEE Signal Process. Lett.*, vol. 26, no. 6, pp. 888–892, Jun. 2019.
- [41] S. Cao and W. Zhang, "Carrier frequency and symbol rate estimation based on cyclic spectrum," *J. Syst. Eng. Electron.*, vol. 31, no. 1, pp. 37–44, Feb. 2020.
- [42] R. C. Gonzalez, R. E. Woods, and S. L. Eddins, *Digital Image Processing Using MATLAB*, 2nd ed. Knoxville, TN, USA: Gatesmark Publ., 2009.
- [43] V. M. Vargas, P. A. Gutiérrez, J. Barbero-Gómez, and C. Hervás-Martínez, "Activation functions for convolutional neural networks: Proposals and experimental study," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 3, pp. 1478–1488, Mar. 2023.
- [44] J. Liu et al., "Toward robust fault identification of complex industrial processes using stacked sparse-denoising autoencoder with softmax classifier," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 428–442, Jan. 2023.
- [45] J. Wang et al., "A novel underwater acoustic signal denoising algorithm for gaussian/non-Gaussian impulsive noise," *IEEE Trans. Veh. Technol.*, vol. 70, no. 1, pp. 429–445, Jan. 2021.
- [46] H.-K. Le, V.-P. Hoang, V.-S. Doan, M.-T. Hoang, and N. P. Dao, "Performance analysis of convolutional neural networks with different window functions for automatic modulation classification," in *Proc. 13th Int. Conf. Commun. Technol. Converg. (ICTC)*, Oct. 2022, pp. 153–157.
- [47] J. Ma, S.-C. Lin, H. Gao, and T. Qiu, "Automatic modulation classification under non-Gaussian noise: A deep residual learning approach," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–6.
- [48] R. Zhang, Y. Zhao, Z. Yin, D. Li, and Z. Wu, "A reference signal-aided deep learning approach for overlapped signals automatic modulation classification," *IEEE Commun. Lett.*, vol. 27, no. 4, pp. 1135–1139, Apr. 2023.
- [49] F. Gardner, "A BPSK/QPSK timing-error detector for sampled receivers," *IEEE Trans. Commun.*, vol. 34, no. 5, pp. 423–429, May 1986.
- [50] M. Bertolucci, R. Cassettari, and L. Fanucci, "CCSDS 131.2-B-1 frequency estimation trade-offs and a novel multi-algorithm FPGA architecture," in *Proc. Int. Conf. Comput., Electron. Commun. Eng. (iCCECE)*, Aug. 2020, pp. 224–229.
- [51] G. Meoni et al., "CCSDS 131.2-B-1 telemetry transmitter: A VHDL IP core and a validation architecture on board RTG4 FPGA," *Acta Astronautica*, vol. 176, pp. 484–493, Nov. 2020.
- [52] P. W. Kinman and J. B. Berner, "Carrier synchronization of offset QPSK for deep space telemetry," in *Proc. IEEE Aerosp. Conf.*, vol. 3, Mar. 2002, p. 3.



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